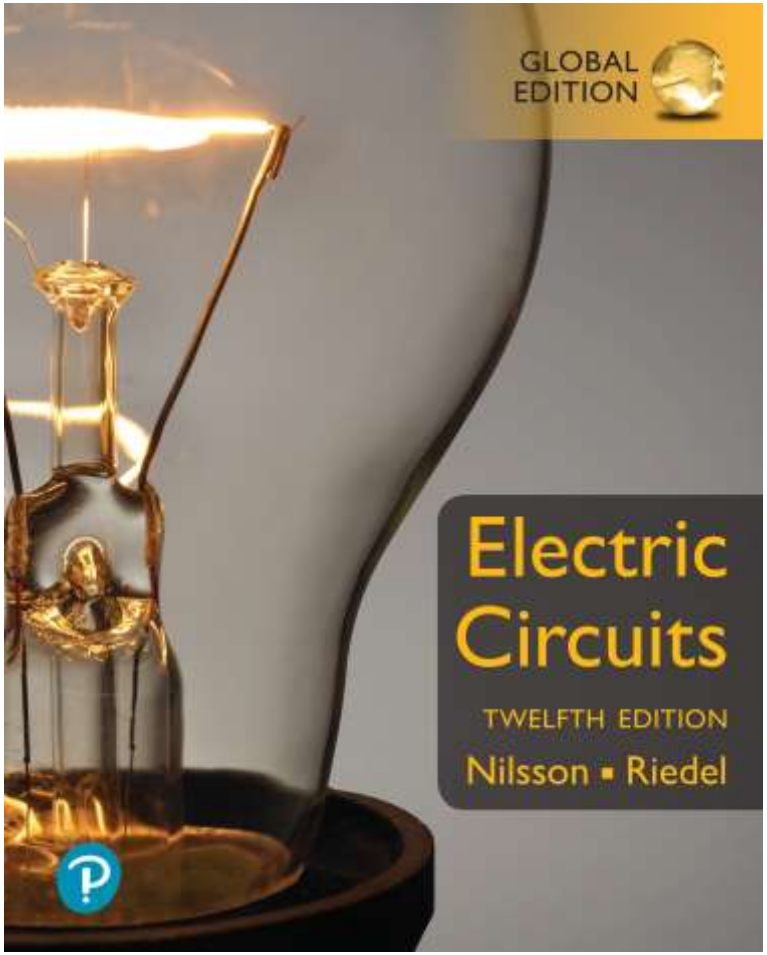
Electric Circuits

Twelfth Edition, Global Edition



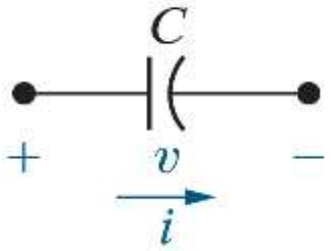
Chapter 6 – Part 2

Inductance, Capacitance, and Mutual Inductance

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Chapter 6 – Capacitors (1 of 16)



- Parallel plates, separated by an insulator, so no charge flows between the plates. Impose a time-varying voltage drop:
  - time-varying electric field
  - time-varying displacement current

- Capacitor equation: $$i(t) = C\frac{dv(t)}{dt}$$

- Units: v(t) is volts, i(t) is amps, and C is farads [F]

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Chapter 6 – Capacitors (2 of 16)

Look at the capacitor equation again:

$$i(t) = C\frac{dv(t)}{dt}$$

Suppose *v(t)* is constant. Then *i(t)* =

A. 0

B. ∞

C. a constant

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Chapter 6 – Capacitors (2 of 16)

Look at the capacitor equation again:

$$i(t) = C\frac{dv(t)}{dt}$$

Suppose *v(t)* is constant. Then *i(t)* =

- ✔ A. 0
- ✘ B. ∞
- ✘ C. a constant

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Chapter 6 – Capacitors (3 of 16)

So, if the voltage drop across the capacitor is constant, its current is 0, so the capacitor can be replaced by

A. a short circuit

B. an open circuit

C. a constant

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Chapter 6 – Capacitors (3 of 16)

So, if the voltage drop across the capacitor is constant, its current is 0, so the capacitor can be replaced by

- ✗ A. a short circuit
- ✓ B. an open circuit
- ✗ C. a constant

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Chapter 6 – Capacitors (6 of 16)

- The equation for voltage in terms of current:

$$i(t) = C\frac{dv(t)}{dt} \quad \Rightarrow \quad i(t)dt = Cdv(t)$$

$$\Rightarrow \quad \int_{t_o}^{t} i(\tau)d\tau = C\int_{v(t_o)}^{v(t)} dx$$

$$\Rightarrow \quad v(t) = \frac{1}{C}\int_{t_o}^{t} i(\tau)d\tau + v(t_o)$$

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Chapter 6 – Capacitors (7 of 16)

- Power and energy

$$p(t) = v(t)i(t) = Cv(t)\frac{dv(t)}{dt}$$

$$p(t) = \frac{dw(t)}{dt} = Cv(t)\frac{dv(t)}{dt}$$

$$\Rightarrow \quad dw(\tau) = Cv(\tau)dv(\tau)$$

$$\Rightarrow \quad \int_0^{w(t)} dx = C\int_0^{v(t)} y(\tau)d\tau$$

$$\Rightarrow \quad w(t) = \tfrac{1}{2}Cv(t)^2$$

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Chapter 6 – Capacitors (8 of 16)

- Capacitors, like inductors, are **energy storage devices!**
  - If the initial voltage drop across the capacitor is non-zero, the capacitor is storing energy.

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Chapter 6 – Capacitors (9 of 16)

Suppose the initial voltage drop across a $4\ \mu F$ capacitor is 10 V. The initial energy stored in the capacitor is

A. 400 μJ

B. 200 μJ

C. 20 μJ

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Chapter 6 – Capacitors (9 of 16)

Suppose the initial voltage drop across a $4\ \mu F$ capacitor is 10 V. The initial energy stored in the capacitor is

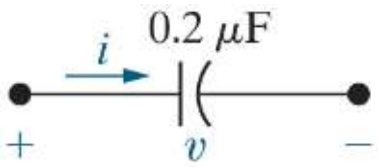
- ✗ A. 400 μJ
- ✓ B. 200 μJ
- ✗ C. 20 μJ

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Chapter 6 – Capacitors (10 of 16)

- Example – Find the voltage, power, and energy



$$i(t)=\begin{cases}0, & t\le 0;\\ 5000t\ \text{A}, & 0\le t\le 20\,\mu\text{s};\\ 0.2-5000t\ \text{A}, & 20\le t\le 40\,\mu\text{s};\\ 0, & t\ge 40\,\mu\text{s}.\end{cases}$$

For $t<0$:

$$v(t)=0\ \text{V};\quad p(t)=0\ \text{W};\quad w(t)=0\ \text{J}$$

For $0\le t\le 20\mu\text{s}$:

$$v(t)=\frac{1}{0.2\mu}\int_0^t 5000x\,dx+v(0)=\frac{1}{0.2\mu}\frac{5000x^2}{2}\bigg|_0^t=12.5\times10^9t^2\ \text{V}$$

$$p(t)=v(t)i(t)=(12.5\times10^9t^2)(5000t)=62.5\times10^{12}t^3\ \text{W}$$

$$w(t)=\tfrac{1}{2}(0.2\mu)(12.5\times10^9t^2)^2=15.625\times10^{12}t^4\ \text{J}$$

At

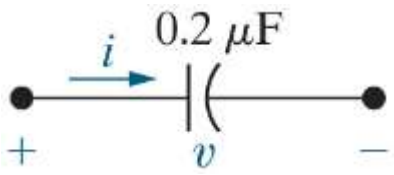
$$t=20\mu\text{s},\qquad v(20\mu\text{s})=12.5\times10^9(20\mu)^2=5\ \text{V}$$

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Chapter 6 – Capacitors (11 of 16)

- Example (continued)



$$i(t)=\begin{cases}0, & t \le 0;\\ 5000t\ \text{A}, & 0 \le t \le 20\,\mu\text{s};\\ 0.2-5000t\ \text{A}, & 20 \le t \le 40\,\mu\text{s};\\ 0, & t \ge 40\,\mu\text{s}.\end{cases}$$

For $20\mu\text{s} \le t \le 40\mu\text{s}$:

$$v(t)=\frac{1}{0.2\mu}\int_{20\mu}^{t}(0.2-5000x)\,dx+v(20\mu)$$

$$=\frac{1}{0.2\mu}\left[0.2x-\frac{5000x^2}{2}\right]\Bigg|_0^t+5$$

$$=(10^6t-12.5\times10^9t^2-10)\ \text{V}$$

$$p(t)=v(t)i(t); \qquad w(t)=\tfrac{1}{2}Cv(t)^2$$

At $t=40\mu\text{s}$,

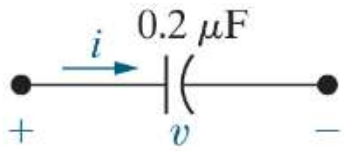
$$v(40\mu\text{s})=[10^6(40\mu)-12.5\times10^9(40\mu)^2-10)=10\ \text{V}$$

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Chapter 6 – Capacitors (12 of 16)

- Example (continued)



$$i(t)=\begin{cases}0, & t\le 0;\\ 5000t\ \text{A}, & 0\le t\le 20\,\mu\text{s};\\ 0.2-5000t\ \text{A}, & 20\le t\le 40\,\mu\text{s};\\ 0, & t\ge 40\,\mu\text{s}.\end{cases}$$

For $t \ge 40\mu s$:

$$v(t)=\frac{1}{0.2\mu}\int_{40\mu}^{t} 0\,dx + v(40\mu)=10\ \text{V}$$

$$p(t)=v(t)i(t)=0\ \text{W}$$

$$w(t)=\tfrac{1}{2}(0.2\mu)(10)^2=10\mu\text{J!}$$

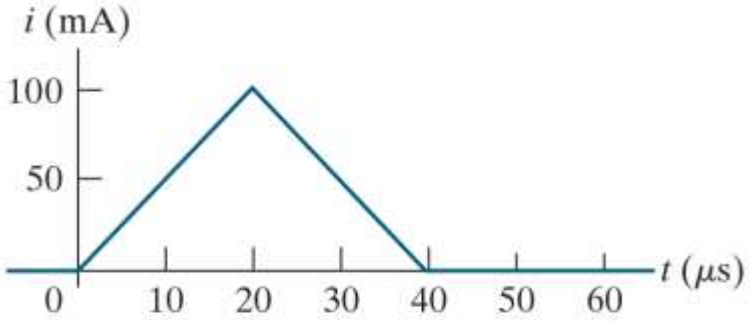
During the interval between 0 and $40\,\mu s$, the power is positive (absorbed), energy is stored and "trapped" by the capacitor, so even when the current goes to 0, the voltage stays at 10 V and the energy is non-zero.

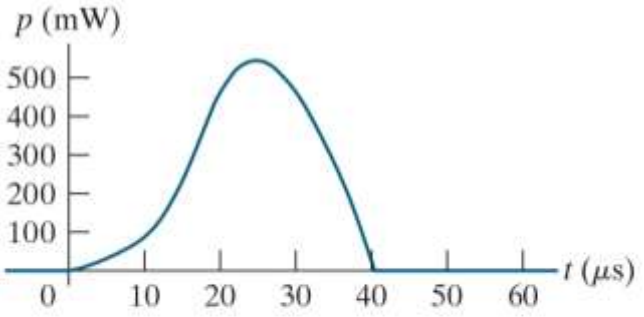
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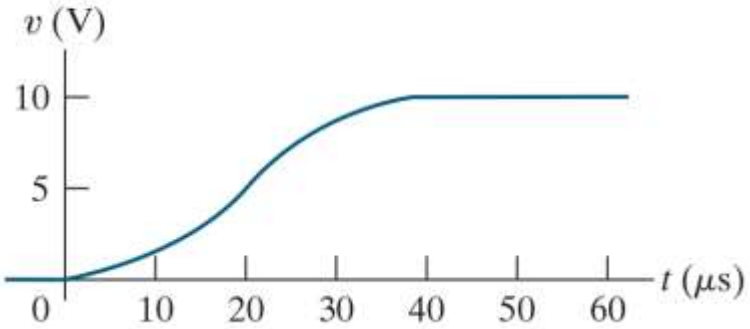
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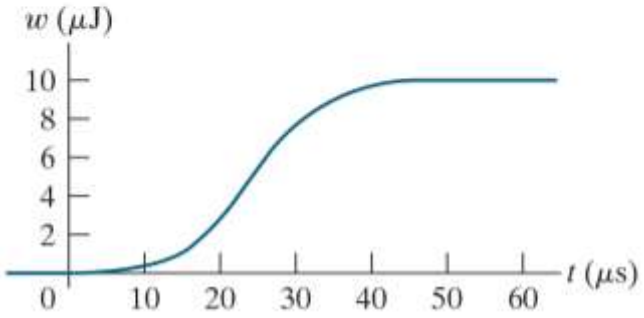
Chapter 6 – Capacitors (13 of 16)

- Example (continued)







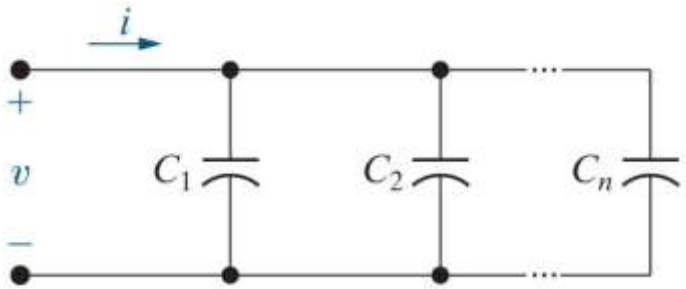


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Chapter 6 – Capacitors (14 of 16)

Capacitors in series



KCL: $$i = i_1 + i_2 + i_3 = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + C_3 \frac{dv}{dt}$$

$$= (C_1 + C_2 + C_3) \frac{dv}{dt} = C_{eq} \frac{dv}{dt}$$

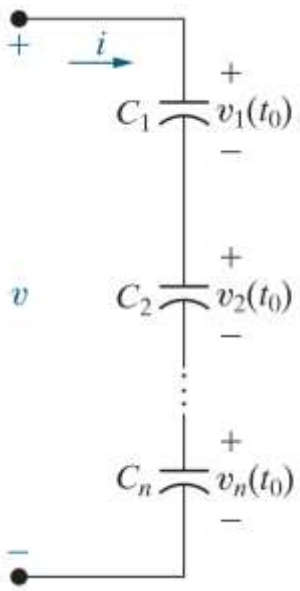
Capacitors in parallel ADD.

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Chapter 6 – Capacitors (15 of 16)

Capacitors in parallel



KVL: $v = v_1 + v_2 + v_3 = \frac{1}{C_1}\int_{t_o}^{t} i(\tau)d\tau + v_1(t_o)$

$$+\frac{1}{C_2}\int_{t_o}^{t} i(\tau)d\tau + v_2(t_o)$$

$$+\frac{1}{C_3}\int_{t_o}^{t} i(\tau)d\tau + v_3(t_o)$$

$$=\left(\frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3}\right)\int_{t_o}^{t} i(\tau)d\tau$$

$$+[v_1(t_o)+v_2(t_o)+v_3(t_o)]$$

$$=\frac{1}{C_{eq}}\int_{t_o}^{t} i(\tau)d\tau + v_{eq}(t_o)$$

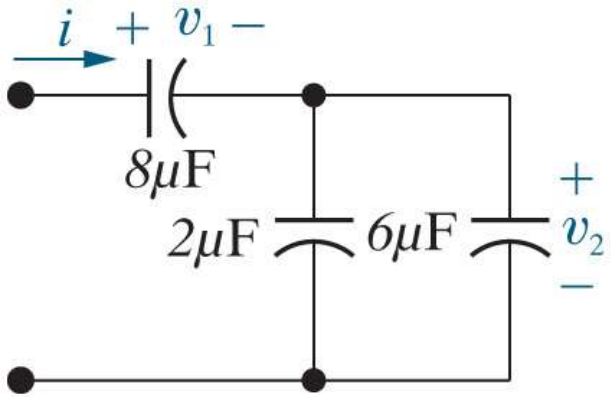
Capacitors combine in series like resistors or inductors combine in parallel.

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Chapter 6 – Capacitors (16 of 16)

Find the equivalent capacitance for the circuit below, assuming $v_1 = 12\,\text{V}$ and $v_2 = -8\,\text{V}$.



A. $C_{eq} = 4\mu\text{F}, v_{eq} = 4\ \text{V}$

B. $C_{eq} = 9.5\mu\text{F}, v_{eq} = 4\ \text{V}$

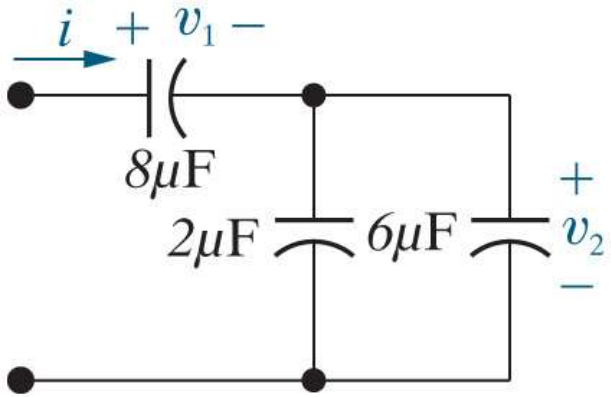
C. $C_{eq} = 4\mu\text{F}, v_{eq} = 20\ \text{V}$

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Chapter 6 – Capacitors (16 of 16)

Find the equivalent capacitance for the circuit below, assuming $v_1 = 12\,\text{V}$ and $v_2 = -8\,\text{V}$.



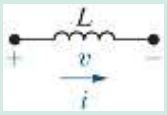
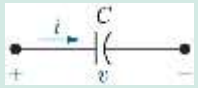
- ✔ A. $C_{eq} = 4\mu\text{F}, v_{eq} = 4\ \text{V}$
- ✘ B. $C_{eq} = 9.5\mu\text{F}, v_{eq} = 4\ \text{V}$
- ✘ C. $C_{eq} = 4\mu\text{F}, v_{eq} = 20\ \text{V}$

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Chapter 6 – Inductance and Capacitance (1 of 2)

Inductor and Capacitor comparison

	Inductor	Capacitor
Symbol		
Units	Henries [H]	Farads [F]
Describing equation	$v(t) = L\dfrac{di(t)}{dt}$	$i(t) = C\dfrac{dv(t)}{dt}$
Other equation	$i(t) = \dfrac{1}{L}\int_{t_o}^{t} v(\tau)d\tau + i(t_o)$	$v(t) = \dfrac{1}{C}\int_{t_o}^{t} i(\tau)d\tau + v(t_o)$
Initial condition	$i(t_o)$	$v(t_o)$
Behavior with const. source	If $i(t) = I$, $v(t) = 0$ $\rightarrow$ short circuit	If $v(t) = V$, $i(t) = 0$ $\rightarrow$ open circuit
Continuity requirement	$i(t)$ is continuous so $v(t)$ is finite	$v(t)$ is continuous so $i(t)$ is finite

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Chapter 6 – Inductance and Capacitance (2 of 2)

Inductor and Capacitor comparison

-	Inductor	Capacitor
Power	$p(t) = v(t)i(t) = Li(t)\frac{di(t)}{dt}$	$p(t) = v(t)i(t) = Cv(t)\frac{dv(t)}{dt}$
Energy	$w(t) = \frac{1}{2}Li(t)^2$	$w(t) = \frac{1}{2}Cv(t)^2$
Initial energy	$w_o(t) = \frac{1}{2}Li(t_o)^2$	$w_o(t) = \frac{1}{2}Cv(t_o)^2$
Trapped energy	$w(\infty) = \frac{1}{2}Li(\infty)^2$	$w(\infty) = \frac{1}{2}Cv(\infty)^2$
Series-connected	$L_{eq} = L_1 + L_2 + L_2$ $i_{eq}(t_o) = i(t_o)$	$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$ $v_{eq}(t_o) = v_1(t_o) + v_2(t_o) + v_3(t_o)$
Parallel-connected	$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3}$ $i_{eq}(t_o) = i_1(t_o) + i_2(t_o) + i_3(t_o)$	$C_{eq} = C_1 + C_2 + C_2$ $v_{eq}(t_o) = v(t_o)$

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